Assignment-1: Tool and API Exploration

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CSAI 810 Section 001 – Group 3

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1. Microsoft LayoutLMv3

Feature	Explanation of Feature
Name of tool / API	**Microsoft LayoutLMv3**
Link	https://github.com/microsoft/unilm/tree/master/layoutlmv3
Tool category (informal)	Multimodal Document Understanding Model / Parser
Purpose	Enables GenAI systems to **understand documents** by jointly learning text, layout, and visual information for tasks like Q&A, form understanding, and layout analysis.
Open-source / Proprietary	**Restricted Open Source**. The code is **MIT/Apache 2.0**, but the pre-trained weights are **CC-BY-NC-SA 4.0** (Non-Commercial). It is **free for students**.
Authors / Organization	**Microsoft Research Asia**
Underlying programming language(s)	**Python**. It is implemented primarily using the **PyTorch** framework and is integrated into the **Hugging Face Transformers library**.
Typical system role	**Document understanding, interpreting text, layout, and visuals.**
API or interaction style	**Python SDK**. Interacted with via the **Hugging Face transformers library.**
Output characteristics	Outputs **precise, layout-aware, and multimodal results**, such as classified text, extracted information, or document layout predictions.
Ease of use	**Moderate**. It requires an external OCR engine to be installed and configured first.
Limitations and constraints	**Sequence Length**: It has a limit of 512 tokens per pass. **License**: The non-commercial weight license is a legal constraint.
Relevance to final project	Critical for the **Parser Agent**. LayoutLM provides structural intelligence that basic PDF libraries lack.

2. Mistral 7B

Feature	Explanation of Feature
Name of tool / API	**Mistral 7B**
Link	https://docs.mistral.ai/
Tool category (informal)	**Foundation Model (LLM) / Reasoning Engine**
Purpose	**High-performance language model** used in GenAI systems for efficient text and code understanding, generation, planning, and reasoning.
Open-source / Proprietary	**Open Source (Apache 2.0)**. The model weights are **fully free** and open for any use. The API service (La Plateforme) is **proprietary but offers a free tier**.
Authors / Organization	**Mistral AI**
Underlying programming language(s)	**Python** (client libraries), **Rust** (underlying inference engine), and **C++.**
Typical system role	**Orchestration / Content Generation / language reasoning**
API or interaction style	**REST API** (cloud) or **Python** Library (local via transformers or llama.cpp).
Output characteristics	Generates accurate, coherent, and **context-aware text or code** efficiently, even for long inputs.
Ease of use	**High**. The API is compatible with OpenAI's client format, making it very easy to swap in. **Local setup is harder but free**.
Limitations and constraints	**Rate Limits**: The free API tier has strict rate limits **(Requests Per Second),** which might slow down batch processing. **Context Window**: 8k to 32k tokens. It may **hallucinate or misprioritize content** without strong prompt constraints or structured input.
Relevance to final project	Essential for the **Planner Agent**. Our project requires "**autonomous decision-making logic**" to summarize chapters. Mistral 7B offers the best balance of reasoning **capability and efficiency** for this project.

3. Suno Bark

Feature	Explanation of Feature
Name of tool / API	**Suno Bark**
Link	https://github.com/suno-ai/bark
Tool category (informal)	**Generative Audio / Text-to-Speech (TTS)**
Purpose	Converts text into realistic speech and audio, enabling GenAI systems to **generate voice, sound effects, and nonverbal sounds** for multimedia applications.
Open-source / Proprietary	**Open Source (MIT License)**. **Fully free** for commercial and non-commercial use.
Authors / Organization	**Suno AI**
Underlying programming language(s)	**Python**. Built on **PyTorch** and uses the **EnCodec neural audio codec**.
Typical system role	**Text-to-audio engine**, generating speech and sounds from text for applications like voice assistants and multimedia content.
API or interaction style	**Python Library**. Interacted with via **Hugging Face transformers (BarkModel)** or the **suno/bark package.**
Output characteristics	Outputs high-quality, expressive, **multilingual audio**, including speech, music, sound effects, and nonverbal sounds.
Ease of use	**Moderate**. It is easy to install, but controlling the output requires **careful prompting**.
Limitations and constraints	**Generation Length**: Limited to **~14 seconds** per pass. We MUST write code to split sentences and stitch audio together. **Hardware**: Requires decent GPU VRAM (or runs slowly on CPU).
Relevance to final project	Suno Bark for text-to-speech makes content **more engaging, accessible, and inclusive** by turning documents into natural, expressive audio that supports diverse learning styles and visually impaired users. And this is what we need for our project.